Code, Data & Decisions

From First Principles to Real-World AI & Data Science

Day 1 & 2 · Orientation and The Big Picture

BS(CS), 6th Semester — University of Bahrain

Day 1 — Where We Begin

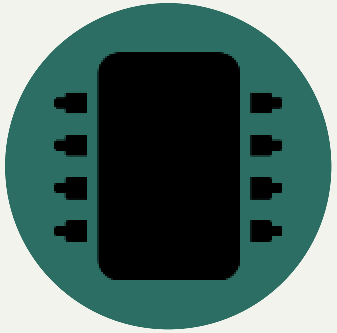
- What is programming?
- How does a computer "think"?
- Compiled vs. interpreted languages
- Why we use Python
- Why this course exists
- Our 16-week roadmap
- Tools to install before next class

Programming = Giving Clear Instructions

"Say exactly what you mean"

- People understand hints and context. Computers do not.
- A program is just a list of simple steps. This list is called an algorithm.
- Try this: tell someone how to make a sandwich, step by step. You will find gaps you never noticed.
- Programming means noticing your own gaps before the computer does.

Inside a Computer: Just Switches



**2 states only:
on / off**

- A computer is made of billions of tiny switches called transistors.
- Each switch is either on or off. We call this 1 and 0.
- A program is like a path — each step moves you to the next point on that path.
- Nothing "understands" your code. That is why we must be exact.

Two Ways to Run Code

Compiled (C++, Java)

Write Code

↓

Compiler Translates It

↓

Computer Runs It Fast

Interpreted (Python)

Write Code

↓

Interpreter Reads Each Line

↓

Runs Right Away

Simple rule: Compiled = runs fast. Interpreted (Python) = lets you test ideas fast.

Why Python? (And Its Limits)

Why Python Is Great

- Reads almost like English
- All the best data tools live here (NumPy, Pandas, scikit-learn)
- A 200-line C++ program can be 20 lines in Python
- Fewer strict rules — easier to turn an idea into working code

Where It Falls Short

- Runs slower than compiled languages
- Not built for parallel processing (a detail called the GIL)
- Not a great fit for mobile apps or embedded devices
- Its speed in data work often comes from C code running underneath

WHY THIS COURSE

"You are not here to become computer scientists or software engineers. You are here so your research never waits on someone else."

And if you enjoy it — this is a real path into data science.

The 16-Week Journey

1 · Orientation

Week 1 — what a computer is, why Python, first code

2 · Python Core

Weeks 2–4 — functions, lists, dicts, files, errors

3 · Stats & Math

Weeks 5–7 — just the math the next phase needs

4 · NumPy, Pandas, Viz

Weeks 8–10 — the real tools of data work

5 · EDA

Weeks 11–12 — reading a real dataset before modeling it

6 · ML + Capstone

Weeks 13–16 — first models, and a project of your own

BEFORE NEXT CLASS

Tools You Will Need

Five simple tools — install them before our next session.

Miniconda

Manages Python and its tools

- Miniconda installs Python on your computer
- It also creates separate, clean workspaces called "environments" for each project
- This stops different projects from breaking each other's tools
- Think of it as a toolbox manager for all your Python projects

Visual Studio Code (VS Code)

Where you write your code

- VS Code is a free, popular code editor
- It highlights your code in color, so mistakes are easy to spot
- It has helpful extensions — small add-ons for Python, Jupyter notebooks, and more
- This is the app you will open every single class

Git — Version Control

"Save points"
for your code

- Git saves snapshots of your code as you work — like checkpoints in a game
- If you break something, you can go back to an earlier snapshot
- It also lets many people work on the same project without overwriting each other
- Git runs on your own computer

GitHub

Your code's online home

- GitHub stores your Git snapshots online, safely
- It lets you share your code, back it up, and show it to others (like future employers)
- You connect Git (on your computer) to GitHub (online) so the two stay in sync
- Almost every professional programmer and researcher uses GitHub

Linux / Ubuntu (Optional)

Optional — not required

- Linux is another operating system, like Windows or macOS
- Ubuntu is a beginner-friendly version of Linux
- Many data science and AI tools were originally built for Linux
- You do NOT need it for this course — Windows or macOS work perfectly fine

Day 1 Task

Set Up Your Coding Environment

- Install Miniconda on your computer
- Install VS Code on your computer
- Install Git on your computer
- Create your own GitHub account
- Connect Git (your computer) to your GitHub account

Due before our next class · Ask in the group chat if you get stuck

Day 1 gave you the foundations.
Day 2 shows you where they lead.

Day 2 — Where This Takes You

- What does "Intelligence" mean?
- What does "Artificial" mean?
- What is Artificial Intelligence?
- A short history of AI
- The main areas of AI
- Real research examples
- What is Data Science?

UNDERSTANDING INTELLIGENCE

What Does "Intelligence" Actually Mean?

Where Does the Word Come From?

There is no single agreed definition of "intelligence" — even experts disagree.

It is a noun, and it comes from the Latin word "Intelliger", which means "to understand."

To Understand, Then To Learn

Intelligence has two connected ideas:

1) To understand something. 2) To learn from it.

We always understand first — and only then can we truly learn.

Capacity to Acquire Capacity

Intelligence is not just what you know — it is how ready you are to gain new abilities.

Ability to Reason

Intelligence is the skill of thinking things through logically, step by step.

Ability to Judge

Intelligence means being able to weigh choices and decide what is right or best.

Ability to Relate Concepts and Situations

Intelligence means connecting what you already know to a brand-new situation.

★ ONE OF THE BEST

Ability to Handle Situations Flexibly

This is often called one of the best definitions of intelligence: the ability to adapt — to bend, not break — when a situation changes. This is also called resilience.

Goal-Oriented Behavior

Intelligence is behavior aimed at a clear goal — adapting your actions until you reach it.

— Robert Sternberg & Salter, Definition of Intelligence

Ability to Take Profit from Experience

Intelligent behavior means learning from your past — using old experience to do better next time.

Proper Use of Limited Resources

Intelligence means using what little time, energy, or resources you have in the smartest possible way to reach your goal.

— Ray Kurzweil (American computer scientist and author), Definition of Intelligence

Getting Better Over Time

One simple, powerful idea: if something keeps improving with practice and experience, that is intelligence.

— Roger Schank (American computer scientist), Definition of Intelligence

What Does "Artificial" Mean?

"Artificial" is an adjective. It comes from the Latin word "Artificium."

Some researchers trace it further back to "Artefact" — meaning something man-made, not natural.

Artificial = Man-Made

A natural lake is made by nature. An artificial lake is built by people.

In the same way, "Artificial Intelligence" means intelligence that is built by people — not born naturally.

Artificial + Intelligence

"Man-made" ability to understand, reason, and learn.

Now let's look at how experts formally define this field.

FORMAL DEFINITIONS

How Do Experts Define Artificial Intelligence?

The General Definition

AI is the ability of a system to understand, learn, gain knowledge, and use that knowledge in the right way.

Buchanan's Definition

AI is the part of computer science that solves problems which humans, right now, can still do better than computers.

Buchanan's Symbolic AI Definition

AI can solve problems by turning information into symbols, then using simple rules of thumb (called heuristics) on those symbols.

Heuristics: are mental shortcuts or "rules of thumb" that allow the brain to solve problems and make quick judgments with minimal effort

Making Computers Smart

In simple words, AI is the area of computer science used to make computers "smart."

Automation of Intelligent Behavior

AI is the branch of computer science that automates behavior we would normally call "intelligent."

Intelligent Agents

AI designs "agents" — programs that can sense their surroundings and take action to reach a goal.

Learning From Experience

AI solves hard problems by learning from experience, and getting better at it over time.

Mimicking Human Behavior

AI is a branch of computer science where machines are taught to mimic human intelligence and behavior.

A SHORT STORY

A Short History of Artificial Intelligence

Al-Jazari (12th Century)

1200s
Early Machines

- One of the earliest inventors of programmable, mechanical machines
- Built automatic devices, water clocks, and human-shaped robots (automata)
- Wrote a famous book: "The Book of Knowledge of Ingenious Mechanical Devices"
- His work laid the groundwork for robotics and automation, centuries early

Alan Turing (1930s–1950s)

1936
The Turing Machine

- A British mathematician who first asked: "Can machines think?"
- Introduced the Turing Machine — the theory behind all modern computers
- In 1950, wrote the paper "Computing Machinery and Intelligence"
- Created the famous Turing Test to check if a machine can act intelligently
- A Turing machine is a simple mathematical model of a computer. Invented by Alan Turing, it explains how any computer problem is solved.

The Dartmouth Conference (1956)

1956
AI Is Born

- A summer research meeting that officially created AI as its own field of study
- Organized by John McCarthy, Marvin Minsky, Claude Shannon, and Nathaniel Rochester
- John McCarthy coined the actual term "Artificial Intelligence" here
- This event marks the true "birthday" of AI as a science

The First AI Winter (1970s)

1970s
Progress Slows

- Interest and funding in AI research dropped sharply
- Computers were still too weak — not enough power or memory
- Many earlier promises about AI turned out to be too optimistic
- Researchers focused more on theory than on working, practical systems

AI Revival (1980s–1990s)

1980s–90s
AI Returns

- Expert Systems became popular and genuinely useful
- Computers became faster and more powerful
- A lot more digital data became available to learn from
- Machine Learning made real, practical progress during this time

THE BUILDING BLOCKS

The Main Areas of Artificial Intelligence

Expert Systems

Rules from human experts

Expert Systems are programs built to copy the decision-making of a real human expert, using stored knowledge and rules.

Used for: medical diagnosis, financial advice, technical troubleshooting, decision support

Machine Learning

Learning from data

Machine Learning lets a computer find patterns in data by itself — instead of a human writing every single rule by hand.

Used for: predictions, recommendation systems, fraud detection, speech recognition

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Machine Learning — 4 Types

Supervised Learning

Learns from labeled examples — each has a known correct answer

Unsupervised Learning

Finds hidden patterns in data with no labels at all

Semi-Supervised Learning

Learns from a mix of labeled and unlabeled data

Reinforcement Learning

Learns by trial and error, earning rewards for good choices

Artificial Neural Networks (ANN)

Inspired by the brain

An Artificial Neural Network is a computing system made of connected "neurons" (nodes), loosely inspired by the human brain.
Each connection has a weight. During training, the network adjusts these weights so its outputs get closer and closer to the correct answer.

Used for: image recognition, speech translation, self-driving cars, chatbots

Evolutionary Algorithms

Inspired by evolution

These methods borrow ideas from natural evolution — testing many possible solutions, and keeping the best ones over many rounds.

Used for: optimization problems, scheduling, engineering design

Computer Vision

Machines that "see"

Computer Vision lets a computer understand and interpret images and videos, much like a human eye and brain would.

Used for: facial recognition, medical imaging, self-driving cars, security cameras

Robotics

AI with a physical body

Robotics combines AI with physical machines, so robots can sense their surroundings, plan, and act on their own.

Used for: manufacturing, healthcare, military, space exploration

Natural Language Processing (NLP)

Machines that "talk"

NLP lets a computer understand, process, and even generate human language — written or spoken.

Used for: chatbots, translation, speech recognition, sentiment analysis, virtual assistants

Swarm Intelligence

Inspired by ants & bees

Swarm Intelligence copies how groups of simple animals — ants, bees, birds, fish — solve problems together, with no single leader.

Used for: optimization, delivery routing, scheduling, distributed problem-solving

AI Ethics

Building AI responsibly

AI Ethics makes sure AI systems are built fairly and safely for everyone — this area is growing quickly as AI spreads.

Used for: fairness, bias, privacy, transparency, accountability, safety, security

AI Fields at a Glance

Expert Systems

Machine Learning

Evolutionary Algorithms

Computer Vision

Robotics

NLP

Swarm Intelligence

AI Ethics

Career Paths in AI

NLP Engineer

Builds chatbots, translators, and language tools

Computer Vision Engineer

Builds systems that understand images & video

Expert System Developer

Builds rule-based decision tools

Machine Learning Engineer

Builds and trains learning models

Optimization Researcher

Works on swarm & evolutionary methods

Robotics Engineer

Builds physical, autonomous machines

REAL RESEARCH

Where This Shows Up in the Real World

How This Course Helps Your Research

Same tools, every field

- Every field now creates more data than people can check by hand
- The tools in this course — Python, statistics, EDA, ML — are used in almost every research field today
- You do not need to become an AI expert. You just need enough skill to ask your data a question, and get a real answer

Case Study: Agriculture

Precision farming

- Predicting crop yield using soil, weather, and satellite data
- Computer Vision spotting plant disease from leaf photos
- Sensors (soil moisture, temperature) automating watering

Case Study: Robotics

Perception + action

- Combining camera, LiDAR, and motion data into one picture of the world
- Path-planning — deciding the best route using probability and math
- Reinforcement Learning — robots improving through trial and error

Case Study: Self-Driving Cars

Millions of miles of data

- Computer Vision spotting lanes, signs, and people in real time
- Huge labeled datasets — the same EDA & cleaning skills from this course apply directly
- Decision-making models trained on real driving data

Case Study: Any Research Project

Faster, stronger work

- Automating data cleaning that used to take days by hand
- Statistical tests to check — or challenge — a hypothesis
- Charts and visuals that make a thesis or paper much stronger

What Is Data Science?

Data → insight → decision

- Data Science means pulling useful insight and decisions out of data
- It combines statistics, programming, and knowledge of a specific field
- AI asks "can a machine act smart?" — Data Science asks "what does this data actually tell us?"
- The two fields overlap constantly — most AI is built and tested using data science methods

Career Paths in Data Science

Data Analyst

Turns raw data into reports and dashboards

Data Engineer

Builds the pipelines that move and store data

Data Scientist

Builds models, makes predictions, runs experiments

ML Engineer

Puts models into real, working products

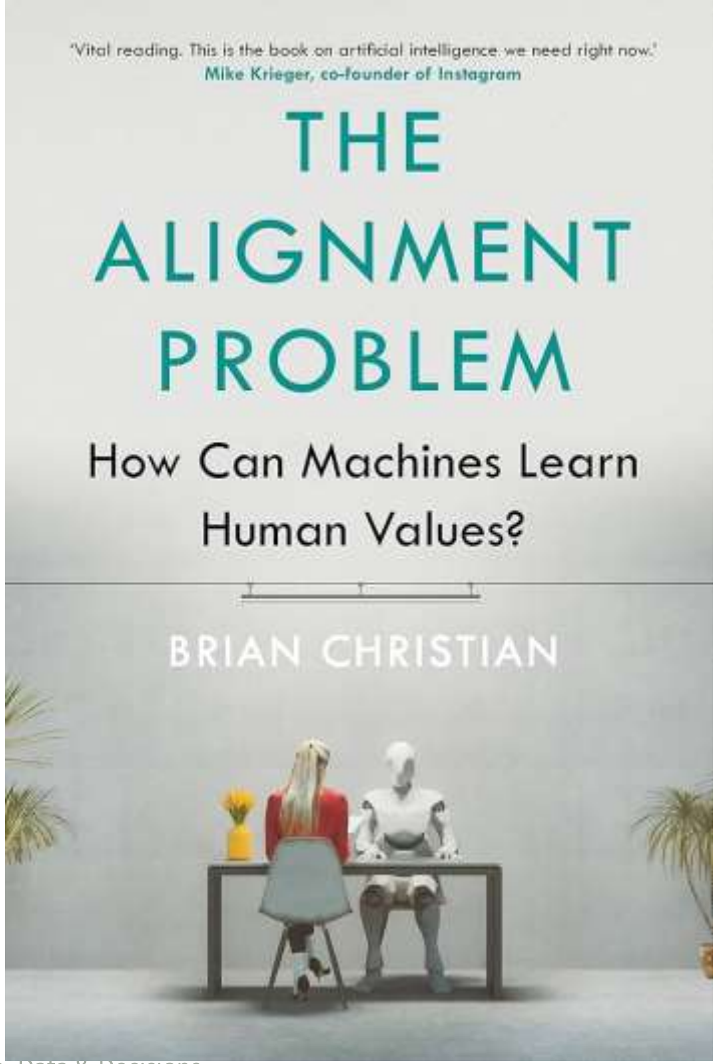
BI Analyst

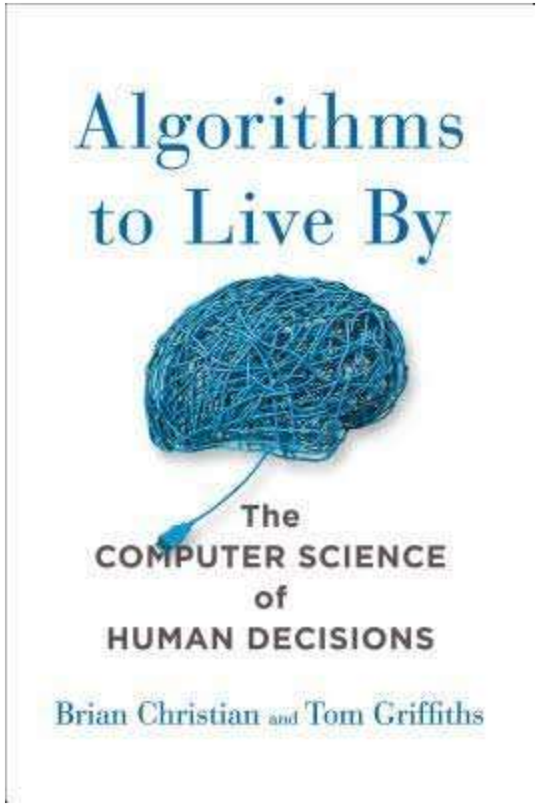
Tracks business metrics to support decisions

Research Scientist

Pushes the field forward in labs or academia

Two interesting Books to understand AI and Algorithms Deeply





A lecture by the Author on the Alignment Problem: https://youtu.be/z6atNBhItBs?si=eFS5ZURNpYF094rC

WHERE YOU FIT IN

Whatever field pulls you — agriculture, robotics, healthcare, or policy — the path starts with the same first step: today.

See you next session.